

Andrea Agiollo

PostDoctoral Researcher in Computer Science and Engineering

🏠 Ecodusweg 47, Delft, 2614WS, The Netherlands

📞 +31 0612583312 ✉ andrea.agiollo@gmail.com

🌐 scholar.google.com 🆔 0000-0003-0531-1978 🌐 github.com

Profile

Postdoctoral researcher with a PhD in Computer Science and Engineering, specializing in the security, privacy, and trustworthiness of Neuro-Symbolic (NeSy) AI systems. Focuses on bridging machine learning with structured logic to architect foundational AI models that are rigorously stress-tested against adversarial vulnerabilities, including membership inference and backdoor attacks, to ensure absolute reliability. Applies these secure, logically grounded, and explainable NeSy agents directly to the domain of cybersecurity. By leveraging structured threat knowledge – such as attack graphs and CVEs – engineers autonomous threat detection systems that enhance the robustness of intrusion detection and risk analysis. A motivated team player with a proven track record of meeting strict deadlines, solving complex problems, and successfully bridging the gap between theoretical AI security and applied systems defense to deliver real-world solutions.

Areas of Expertise

Machine Learning - Neuro-Symbolic AI - AI Trustworthiness - Cybersecurity - Resource Efficiency - eXplainable AI

Current Position

PostDoc at Cybersecurity Group at EEMCS Faculty [TU Delft](#)

Delft, NL 10/2024 - Ongoing

My current research advances the foundational security and robustness of Neuro-Symbolic (NeSy) AI agents, systematically stress-testing these hybrid models against sophisticated adversarial threats, including membership inference and backdoor attacks. Rather than merely evaluating vulnerabilities, I engineer resilient architectures that harden the integration of neural and symbolic reasoning. Concurrently, I leverage these trustworthy NeSy agents to architect highly robust intrusion detection and threat analysis systems. By grounding these autonomous agents in structured threat intelligence – such as formal attack definitions and attack graphs – I deliver defense mechanisms that ensure rigorous interpretability and high detection accuracy in complex network environments.

Education

Ph.D. in Computer Science and Engineering [University of Bologna](#)

Bologna, IT 11/2020 - 06/2024

Thesis entitled “*Embedding AI into Constrained Devices: A Multi-Faceted Approach*”

MSc in ICT for Internet and Multimedia [University of Padua](#)

Padua, IT 09/2018 - 07/2020

Graduation Grade: 110/110 cum laude

MSc in Communication Engineering [National Taiwan University](#)

Taipei, TWN 09/2018 - 07/2020

Graduation Grade: 4.23/4.3

BSc in Information Engineering [University of Padua](#)

Padua, IT 10/2015 - 07/2018

Graduation Grade: 110/110 cum laude

Erasmus Exchange [University of Aberdeen](#)

Aberdeen, Scotland, UK 09/2017 - 01/2018

Relevant Activities

Organization:

Web-Chair at ECAI 2025

Bologna, Italy

I served as the Web Chair for ECAI 2025, the premier European conference on Artificial Intelligence, where I lead the design, development, and management of the conference website.

Organizing Committee Member of the 1st HotDiSec Workshop

Rome, Italy

To be held in conjunction with ESORICS 2026. The HotDiSec workshop provides a platform for researchers and practitioners working in the distributed ML and security domain to share high-impact research, rigorous critiques, and forward-looking discussion papers.

Organizing Committee Member of the 1st ANSyA Workshop

Bologna, Italy

Held in conjunction with ECAI 2025, the workshop's first edition was a success, gathering 30 submissions. The ANSyA workshop provides a platform for applied neuro-symbolic systems engineering, fostering collaborations between academia and industry, and encouraging innovative thinking. ANSyA revitalized the applied dimension of neuro-symbolic AI, previously underestimated.

Organizing Committee Member of the [1st HotDiML Workshop](#)

Glasgow, Scotland, UK

Held in conjunction with ICDCS 2025, the workshop's first edition was a success, gathering *11 submissions*. HotDiML serves as a platform for critical thinking, sharing both successes and failures, and fostering constructive discussions about distributed machine learning applications. HotDiML encourages novel directions, accepting papers that offer fresh perspectives or critical analyses, even if not fully developed.

Publicity Chair of the [6th EXTRAAMAS Workshop](#)

Auckland, New Zealand

Held in conjunction with AAMAS 2024. I served as the Publicity Chair for the EXTRAAMAS 2024 workshop, one of the premier venues on the explainability and transparency of intelligent agent systems.

Patents and Inventions:

Patent: "Dishwasher for Treating Washware"

Pordenone, IT

- *Authors:* Andrea Comina, Andrea Agiollo, and Andrea Omicini.
- *Status:* Approved on September 10th 2024 from the European Patent Office and valid worldwide.
- *Full access:* <https://patentimages.storage.googleapis.com/56/9e/84/4f5f1e694a9f3c/EP4427656A1.pdf>
- Done as part of my PhD program in collaboration with Electrolux Professional

Awards and Honors:

- 🏆 **Gold Reviewer** *43rd International Conference on Machine Learning ([ICML-26](#))*
- 🏆 **Outstanding Reviewer** *42nd International Conference on Machine Learning ([ICML-25](#))*
- 🏆 **Outstanding Reviewer** *31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining ([KDD-25](#))*
- 🏆 **Top Reviewer** *39th Conference on Uncertainty in Artificial Intelligence ([UAI-23](#))*
- 🏆 **Student Award** *Double Degree Honoree for Master Degree Program at University of Padua (2020)*

Appointments:

Visiting Research Scholar at Purdue University

Lafayette, Indiana, USA 08/2023 - 11/2023

My research efforts focused on the intersection of neuro-symbolic integration techniques and the properties of neural network learning process, focusing on memorization, minimum flatness and many others. Throughout this experience, I had the pleasure of collaborating with an international research group, strengthening my knowledge of the inner properties of sub-symbolic models.

Visiting Research Scholar at Delft University of Technology

Delft, NL 10/2022 - 02/2023

My research efforts focused on eXplainable Artificial Intelligence (XAI) techniques to extract knowledge from a trained Large Language model used for text classification and morality alignment. Throughout this experience, I had the pleasure to collaborate with an international research group, strengthening my knowledge of XAI and Natural Language Processing.

Research Community Service:

- PC Member** *40th Annual Conference on Neural Information Processing Systems ([NeurIPS-26](#))*
- PC Member** *32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining ([KDD-26](#))*
- PC Member** *43rd International Conference on Machine Learning ([ICML-26](#))*
- PC Member** *42nd Conference on Uncertainty in Artificial Intelligence ([UAI-26](#))*
- PC Member** *40th AAAI Conference on Artificial Intelligence ([AAAI-26](#))*
- PC Member** *14th International Conference on Learning Representations ([ICLR-26](#))*
- PC Member** *16th ACM Conference on Data and Application Security and Privacy ([CODASPY-26](#))*
- PC Member** *Transactions on Machine Learning Research ([TMLR-26](#))*
- PC Member** *30th European Symposium on Research in Computer Security ([ESORICS-25](#))*
- PC Member** *15th ACM Conference on Data and Application Security and Privacy ([CODASPY-25](#))*
- PC Member** *42nd International Conference on Machine Learning ([ICML-25](#))*
- PC Member** *39th Annual Conference on Neural Information Processing Systems ([NeurIPS-25](#))*
- PC Member** *31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining ([KDD-25](#))*
- PC Member** *39th AAAI Conference on Artificial Intelligence ([AAAI-25](#))*
- PC Member** *Transactions on Machine Learning Research ([TMLR-25](#))*

PC Member *Workshop on EXplainable and TRAnsparent AI and Multi-Agent Systems* ([EXTRAAMAS 2025](#))

PC Member *Applications and Systems for Healthcare* ([ASH 2025](#)) *Track of SAC 2025*

PC Member *30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* ([KDD-24](#))

PC Member *40th Conference on Uncertainty in Artificial Intelligence* ([UAI-24](#))

PC Member *38th AAAI Conference on Artificial Intelligence* ([AAAI-24](#))

PC Member *Workshop on EXplainable and TRAnsparent AI and Multi-Agent Systems* ([EXTRAAMAS 2024](#))

PC Member *Workshop from Objects to Agents* ([WOA 2024](#))

PC Member *39th Conference on Uncertainty in Artificial Intelligence* ([UAI-23](#))

PC Member *37th AAAI Conference on Artificial Intelligence* ([AAAI-23](#))

PC Member *Workshop on EXplainable and TRAnsparent AI and Multi-Agent Systems* ([EXTRAAMAS 2023](#))

PC Member *Workshop from Objects to Agents* ([WOA 2023](#))

PC Member *Workshop from Objects to Agents* ([WOA 2022](#))

External/Invited Reviewer *IEEE International Conference on Computer Communications* ([INFOCOM 2025](#))

External/Invited Reviewer *Elsevier Computer Networks*

External/Invited Reviewer *Elsevier Knowledge Based Systems*

External/Invited Reviewer *IEEE Transactions on Services Computing*

External/Invited Reviewer *IEEE Transactions on Network and Service Management*

External/Invited Reviewer *ACM Transactions on Artificial Intelligence Security and Privacy*

Publications

Bibliometrics: 430 citations, *h-index*: 11, *i10-index*: 14.

Selected Publications:

- Young In Kim, Andrea Agiollo, Pratiksha Agrawal, Johannes O. Royset, and Rajiv Khanna. “On the Privacy Risks of Sharpness Aware Minimization”. In: *International Conference on Learning Representations (ICLR)*. 2026
- Giovanni Ciatto, Andrea Agiollo, Matteo Magnini, and Andrea Omicini. “Large language models as oracles for instantiating ontologies with domain-specific knowledge”. In: *Knowledge-Based Systems* 310 (2025), p. 112940
- Andrea Agiollo, Young In Kim, and Rajiv Khanna. “Approximating Memorization Using Loss Surface Geometry for Dataset Pruning and Summarization”. In: *SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*. 2024
- Giovanni Ciatto, Federico Sabbatini, Andrea Agiollo, Matteo Magnini, and Andrea Omicini. “Symbolic Knowledge Extraction and Injection with Sub-symbolic Predictors: A Systematic Literature Review”. In: *ACM Computing Surveys* 56.6 (June 2024), 161:1–161:35. DOI: 10.1145/3645103
- Andrea Agiollo, Luciano Cavalcante Siebert, Pradeep Murukannaiah, and Andrea Omicini. “From Large Language Models to Small Logic Programs: Building Global Explanations from Disagreeing Local Post-hoc Explainers”. In: *Autonomous Agents and Multi-Agent Systems* (2024)
- Andrea Agiollo, Paolo Bellavista, Matteo Mendula, and Andrea Omicini. “EneA-FL: Energy-aware Orchestration for Serverless Federated Learning”. In: *Future Generation Computer Systems* 154 (May 2024), pp. 219–234. DOI: 10.1016/j.future.2024.01.007
- Andrea Agiollo, Enkeleda Bardhi, Mauro Conti, Riccardo Lazzeretti, Eleonora Losiouk, and Andrea Omicini. “GNN4IFA: Interest Flooding Attack Detection With Graph Neural Networks”. In: *2023 IEEE 8th European Symposium on Security and Privacy (EuroS&P)*. Delft, Netherlands, July 2023, pp. 615–630. DOI: 10.1109/EuroSP57164.2023.00043
- Andrea Agiollo, Enkeleda Bardhi, Mauro Conti, Nicolò Dal Fabbro, and Riccardo Lazzeretti. “Anonymous Federated Learning via Named-Data Networking”. In: *Future Generation Computer Systems* 152 (2024), pp. 288–303. DOI: 10.1016/j.future.2023.11.009
- Andrea Agiollo and Andrea Omicini. “Measuring Trustworthiness in Neuro-Symbolic Integration”. In: *Proceedings of the 18th Conference on Computer Science and Intelligence Systems*. Vol. 35. Annals of Computer Sciences and Information Systems. Sept. 2023, pp. 1–10. DOI: 10.15439/2023F6019

- Andrea Agiollo, Andrea Rafanelli, Matteo Magnini, Giovanni Ciatto, and Andrea Omicini. "Symbolic Knowledge Injection meets Intelligent Agents: QoS metrics and experiments". In: *Autonomous Agents and Multi-Agent Systems* 37.2 (June 2023), 27:1–27:30. DOI: 10.1007/s10458-023-09609-6
- Andrea Agiollo and Andrea Omicini. "GNN2GNN: Graph Neural Networks to Generate Neural Networks". In: *Proceedings of the 38th Conference on Uncertainty in Artificial Intelligence, UAI 2022*. Aug. 2022, pp. 32–42. URL: <https://proceedings.mlr.press/v180/agiollo22a.html>
- Andrea Agiollo, Mauro Conti, Pallavi Kaliyar, TsungNan Lin, and Luca Pajola. "DETONAR: Detection of Routing Attacks in RPL-based IoT". in: *IEEE Transactions on Network and Service Management* 18.2 (2021), pp. 1178–1190. DOI: 10.1109/TNSM.2021.3075496

Other Publications:

- Andrea Agiollo, Roberta Calegari, Giovanni Ciatto, Matteo Magnini, Andrea Omicini, and Federico Sabbatini. "Intelligent agents from symbolic to neurosymbolic systems: the quest for integration". In: *The Agents Journey: Twenty-Five Years of Multi-agent Systems*. Springer, 2026, pp. 320–339
- Andrea Agiollo, Luciano C. Siebert, Pradeep K. Murukannaiah, and Andrea Omicini. "The Quarrel of Local Post-hoc Explainers for Moral Values Classification in Natural Language Processing". In: *Explainable and Transparent AI and Multi-Agent Systems. Fifth International Workshop, EXTRAAMAS 2023, London, UK, May 29, 2023*. 2023
- Mattia Passeri, Andrea Agiollo, and Andrea Omicini. "Peer-Reviewed Federated Learning". In: *WOA 2023 – 24th Workshop "From Objects to Agents"*. Ed. by Rino Falcone, Cristiano Castelfranchi, Alessandro Sapienza, and Filippo Cantucci. Vol. 3579. CEUR Workshop Proceedings. Sun SITE Central Europe, RWTH Aachen University, Nov. 2023, pp. 49–65
- Andrea Agiollo, Andrea Rafanelli, and Andrea Omicini. "Towards Quality-of-Service Metrics for Symbolic Knowledge Injection". In: *WOA 2022 – 23rd Workshop "From Objects to Agents"*. 2022
- Andrea Agiollo, Giovanni Ciatto, and Andrea Omicini. "Shallow2Deep: Restraining Neural Networks Opacity through Neural Architecture Search". In: *Explainable and Transparent AI and Multi-Agent Systems. Third International Workshop, EXTRAAMAS 2021, Virtual Event, May 3–7, 2021, Revised Selected Papers*. Cham, Switzerland, 2021, pp. 63–82. URL: http://link.springer.com/10.1007/978-3-030-82017-6_5
- Andrea Agiollo, Giovanni Ciatto, and Andrea Omicini. "Graph Neural Networks as the Copula Mundi between Logic and Machine Learning: A Roadmap". In: *WOA 2021 – 22nd Workshop "From Objects to Agents"*. Ed. by Roberta Calegari, Giovanni Ciatto, Enrico Denti, Andrea Omicini, and Giovanni Sartor. 2021, pp. 98–115. URL: <http://ceur-ws.org/Vol-2963/paper18.pdf>
- Andrea Agiollo and Andrea Omicini. "Load Classification: A Case Study for Applying Neural Networks in Hyper-Constrained Embedded Devices". In: *Applied Sciences* 11.24 (Dec. 2021). Special Issue "Artificial Intelligence and Data Engineering in Engineering Applications". DOI: 10.3390/app112411957
- Umberto Michieli, Maria Camporese, Andrea Agiollo, Giampaolo Pagnutti, and Pietro Zanuttigh. "Region Merging Driven by Deep Learning for RGB-D Segmentation and Labeling". In: *13th International Conference on Distributed Smart Cameras (ICDSC2019)*. 2019, 9:1–9:6. URL: <https://doi.org/10.1145/3349801.3349810>

Submitted:

- De Filippo L., Bardhi E., Agiollo A., Palma, A., Bonomi, S. & Kuipers, F. Beyond the Syntax: Do Security Experts Trust LLMs for NIDS Rule Engineering? *ACM Conference on Computer and Communications Security (CCS 2026)*.
- Agiollo A., Bardhi, E., Palma, A., Lazzeretti, R., Bonomi, S. & Kuipers, F. SoK: Harmonising Attack Graphs and Intrusion Detection Systems. *ACM Transactions on Privacy and Security*.
- Agiollo A., Young, I. K., & Khanna, R. SAMOSA: Sharpness Aware Minimization for Open Set Active learning. *SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2026)*.
- Agiollo A., & Omicini, A. Integrating Machine Learning into Belief-Desire-Intention Agents: Current Advances and Open Challenges. *ACM Computing Surveys*.

Teaching, Student Supervision & Seminars

Student Supervision Activities:

- Master Thesis Supervisor for student Raoul Saad working on thesis entitled “Towards Neuro-Symbolic AI in Network Intrusion Detection Systems” - A.Y. 2025/2026.
- Master Thesis Supervisor for student Sofia Burkow working on thesis entitled “DeepChronos: Neuro-Symbolic Reasoning for Multi-Step Attack Detection” - A.Y. 2025/2026.
- Master Thesis Supervisor for student Anna Visman working on thesis entitled “A Neuro-Symbolic Framework for Alert Fatigue Reduction in Security Operation Center” - A.Y. 2025/2026.
- Master Thesis Supervisor for student Davide Alpi working on thesis entitled “Conversational Non-Player Characters: A Study on the Derailment of Large Language Models” - A.Y. 2022/2023.
- CSE3000 Research Project (Bachelor Thesis) Supervisor involving 5 Bachelor students – A.Y. 2024/2025.
- Supervisor of Hacking Lab Course Project involving 5 Master students - A.Y. 2024/2025.

Teaching Activities:

- Lecture: “Introduction to Natural Language Processing”; Course: Engineering Intelligent Systems; Venue: University of Bologna (05/2023).
- Teacher Assistant (Tutor) for the “Fondamenti di Informatica” course – A.Y. 2022/2023.
- Teacher Assistant (Tutor) for the “Fondamenti di Informatica” course – A.Y. 2021/2022.

Seminars & Invited Talks:

- Invited Talk: “The Best of Both Worlds: Fixing Data-Driven NIDS with Neuro-Symbolic Logic”; Venue: TU Eindhoven (03/2026).
- Invited Talk: “On the Quest for Resource-Efficient AI: Lessons Learnt from Machine Learning and Neuro-Symbolic Integration”; Venue: Vrije Universiteit Amsterdam (04/2025).
- Seminar: “Neuro-Symbolic AI: The What and Why”; Venue: TU Delft (06/2025).
- Seminar: “Resource-Friendly Artificial Intelligence”; Venue: Delft University of Technology (01/2023).
- Seminar: “Graph Neural Networks - Theory & Applications”; Venue: University of Bologna (04/2021).

Skills

Programming Languages:

- *Advanced:* python, java, C++, LaTeX
- *Intermediate:* C, C#, MATLAB, Kotlin, Bash Linux
- *Basic:* HTML, JavaScript, Prolog

ML/DL Frameworks:

- *Advanced:* Pytorch, TensorFlow, Scikit-Learn
- *Intermediate:* HuggingFace, Keras
- *Basic:* DL4J, ONNX

Development Tools:

- *Versioning:* git
- *Testing:* Unittest, JUnit
- *Others:* IntelliJ, PyCharm, Visual Studio

Operating Systems:

- *Desktop:* MacOS, Windows, Linux
- *Mobile:* Android, iOS

Languages

- **Italian** [Native]
- **English** [Proficient] - C1
- **Spanish** [Basic] - A1